

Toward $P \neq NP$ via Time-Bounded Kolmogorov Complexity (Revised)

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Note on this revision. The author used Claude (Anthropic) as an aid in checking the paper’s citations against the current literature and in critically reviewing the technical arguments of an earlier draft. That review identified a scaling problem in the direct-product amplification argument of Section 5, and an incompleteness in the implication chain of Section 8 (the equivalence of Kabanets and Kolokolova is a conjunction of two conditions, not one). Both sections have been rewritten below to state these issues explicitly rather than to paper over them. All mathematical formulations, the overall research direction, and any remaining errors are the author’s own responsibility.

Abstract

We formulate a concrete research program for separating P from NP through time-bounded Kolmogorov complexity. The approach combines two recent developments: exact complexity-theoretic characterizations of chain rules for time-bounded Kolmogorov complexity and unconditional failures of symmetry of information for randomized time-bounded Kolmogorov complexity. We propose weakened theorems as the central targets: a local, gap-preserving transfer from randomized complexity rK^t to deterministic K^t , and a reduction from SAT through a restricted minimum-circuit-size problem to a mild-gap conditional K^t problem. This revision makes explicit two structural requirements that emerged from a technical review of the original draft. First, the gap required from the transfer theorem must in fact be *linear* in the instance length, not merely superlogarithmic, since the natural direct-product amplification argument cannot bridge that distance; we isolate the one-sided (lower-bound-only) form of the transfer as the more tractable target. Second, the implication from a chain-rule violation to $P \neq NP$ requires, in addition to NP-hardness of the associated meta-complexity problem, an auxiliary circuit-hardness hypothesis on E that is not automatically discharged. With these corrections in place, the program consists of three open bridges rather than two, each of independent interest. A closing outlook (Section 12) calibrates the difficulty of each bridge against known barriers in the hardness-versus-randomness and meta-complexity literature, rather than treating the corrections above as merely cosmetic.

1 Introduction

The P versus NP problem asks whether every problem whose solutions can be verified in polynomial time can also be solved in polynomial time:

$$P \stackrel{?}{=} NP.$$

Since SAT is NP -complete, it is enough to establish

$$SAT \notin P.$$

A classical idea due to Kolmogorov is to approach this question through time-bounded Kolmogorov complexity. For a time bound t , let

$$K^t(x \mid y)$$

denote the length of the shortest program which, given y , outputs x within t computation steps.

Recent work gives this approach a much more precise foundation. Kabanets and Kolokolova established exact complexity-theoretic characterizations for natural chain rules for time-bounded Kolmogorov complexity, tying the validity of a sublinear-error chain rule to a precise pair of complexity-theoretic conditions (made explicit in Section 2 below).

Independently, Hu, Manor, and Oliveira proved an unconditional failure of symmetry of information for randomized time-bounded Kolmogorov complexity rKt , together with strong lower bounds for estimating conditional rKt .

These results suggest a concrete route toward $P \neq NP$ — one that, as this revision clarifies, is more demanding than it first appears.

2 Computational Information

For strings $x_1, \dots, x_\ell, y$, define

$$N = \sum_{i=1}^{\ell} |x_i|.$$

The relevant chain rule has the form

$$K^t(x_1, \dots, x_\ell \mid y) \geq \sum_{i=1}^{\ell} K^{t^c}(x_i \mid y, x_1, \dots, x_{i-1}) - \ell O(\log t) - e(N, t). \quad (1)$$

The important regime is

$$e(N, t) = o(N).$$

Kabanets and Kolokolova show that this type of chain rule is tied to the worst-case complexity of gap versions of minimum conditional time-bounded Kolmogorov complexity. Precisely, the Chain Rule with error $e(N, t) \leq o(N)$ is equivalent to the *conjunction* of two conditions:

- (a) $E \not\subseteq \text{i.o.-SIZE}[2^{o(n)}]$ — a widely conjectured, but currently unproven, circuit lower bound for the exponential-time class E ; and
- (b) $\text{GapMcK}^t P \in \text{promise-}P$, where $\text{GapMcK}^t P$ is the promise problem of distinguishing instances $(x, y, 1^s)$ with $K^t(x \mid y) \leq s$ from those with $K^{\text{poly}(t)}(x \mid y) > s + o(|x|)$.

Under the further hypothesis that $\text{GapMcK}^t P$ is NP-hard — currently known only conditionally, under cryptographic assumptions — condition (b) becomes equivalent to $P = NP$, and the chain rule with sublinear error becomes equivalent to the conjunction of (a) and $P = NP$.

This conjunctive structure matters. Consequently, to prove $P \neq NP$ by this route it is *not* enough to exhibit a family violating (1) by a sufficiently large gap: one must also account for condition (a), since a violation of the chain rule only certifies that (a) or (b) fails, not which one. We return to this point in Section 8, where it is the source of a genuine gap in the original argument.

3 Randomized Complexity as an Unconditional Source of Separation

Let rKt denote the randomized analogue of Levin’s time-bounded Kolmogorov complexity.

For ordinary Kolmogorov complexity, symmetry of information gives

$$K(x, y) = K(x) + K(y \mid x) + O(\log n).$$

For time-bounded complexity this identity becomes substantially more delicate. Deterministically, symmetry of information for K^t is already known to fail unconditionally, at a modest (polylogarithmic) scale, by a classical brute-force argument of Ronneburger [4]; this is what allows the classical Longpré–Watanabe connection between such failures and $NP \not\subseteq P$ to be stated at all, and it is the reason that only a *sufficiently large* (in fact linear) failure, rather than failure per se, can be informative — Ronneburger’s modest failure does not by itself resolve P versus NP .

Hu, Manor, and Oliveira proved in 2026 that symmetry of information fails unconditionally for rKt as well. They also obtained a near-optimal lower bound for estimating conditional rKt , and connect the average-case version of this failure most directly to the existence of one-way functions rather than to NP versus P directly.

Thus randomized time-bounded complexity already contains an unconditional computational-information obstruction. The central question — and the harder of the two bridges this program requires — is whether this obstruction can be transferred to deterministic K^t , and with a large enough gap to matter.

4 Target I: A Local Transfer Theorem

A global equivalence between rKt and K^t is unnecessarily strong. We propose instead a local transfer theorem for explicitly constructed instances.

Theorem Target 1 (Local Gap-Preserving Transfer). *There exists a polynomial-time transformation*

$$T : (x, y, s) \mapsto (x', y', s')$$

and a polynomial q such that, for a designated efficiently constructible family of instances,

$$rKt(x \mid y) \leq s$$

implies

$$K^{q(t)}(x' \mid y') \leq s' + O(\log n), \tag{2}$$

whereas

$$rKt(x \mid y) > s + \Delta(n)$$

implies

$$K^{q(t)}(x' \mid y') > s' + \Omega(\Delta(n)). \tag{3}$$

Remark (Asymmetry between the two directions). *Implications (2) and (3) are not of the same character, and it is worth separating them explicitly.*

Implication (3) is a lower-bound transfer: a pair certified as rKt -hard remains hard, up to the loss $\Delta(n)$, for the deterministic measure K^t . Transfers of this kind are typically amenable to counting or incompressibility arguments, which do not require exhibiting an explicit efficient

procedure, only ruling out the existence of a short deterministic description. This is the more promising target for initial attack.

Implication (2), by contrast, is a constructive transfer: given a short randomized description of x , it demands producing a comparably short deterministic one. This is precisely the kind of step that Hu, Manor, and Oliveira identify as the obstruction to adapting Ronneburger’s unconditional argument for K^t to the randomized measure rKt in the first place — the absence of a pseudodeterministic (canonical, high-success-probability) output for the relevant construction. Known pseudodeterministic constructions of high-complexity strings are currently limited to subexponential running time [6, 5], short of what implication (2) would need here. A one-sided version of Theorem Target I, keeping only implication (3), sidesteps this obstruction and is the version we recommend attacking first; it already suffices for the corrected amplification discussion of Section 5.

The desirable regime for $\Delta(n)$ is discussed, and revised, in Section 5.

5 Gap Amplification — and a Scaling Obstruction

A natural hope is that a mild per-instance gap $\Delta(n)$ from Theorem Target I can be amplified by taking m independent, structured copies

$$X = (x_1, \dots, x_m), \quad Y = (y_1, \dots, y_m),$$

of total length $N = mn$, via a direct-product inequality

$$\text{Gap}(X, Y) \geq m\Delta(n) - O(m \log n). \quad (4)$$

This hope does not survive close inspection, and the original version of this section overstated what such an inequality can deliver. Recall from Section 2 that refuting the Chain Rule for *every* admissible error function $e(N, t) = o(N)$ requires a gap that is $\Omega(N)$ — linear in the *total* length of the instance, not merely large in absolute terms. In the direct-product construction, $N = mn$, so what is actually required is

$$\text{Gap}(X, Y) = \Omega(mn).$$

Equation (4) gives $\text{Gap}(X, Y) = \Theta(m\Delta(n))$ whenever $\Delta(n) = \omega(\log n)$. Comparing the two:

$$\Theta(m\Delta(n)) = \Omega(mn) \iff \Delta(n) = \Omega(n).$$

That is, the direct-product construction converts a per-instance gap into a *total* gap that is an $\Omega(N)$ -fraction of N if and only if $\Delta(n)$ was *already* linear in n . If $\Delta(n) = o(n)$ — in particular if $\Delta(n)$ is merely $\omega(\log n)$ or $\omega(\log \log n)$, as the original Section 5 hoped — then $\Delta(n)/n \rightarrow 0$, and so does $m\Delta(n)/(mn)$, for every choice of polynomial $m(n)$. Concatenating independent copies preserves the *ratio* of gap to length; it does not amplify it. This is the same phenomenon that makes naive repetition insufficient for hardness amplification in circuit complexity, where genuine amplification (e.g., Yao’s XOR lemma) requires a non-trivial combiner rather than a direct product.

Two consequences follow.

First, if Theorem Target I is established only in the sublinear regime $\Delta(n) = \omega(\log n)$, the direct-product construction above does *not* yield a violation of the sublinear-error chain rule. Closing this requires either (i) a fundamentally different combiner — analogous to an XOR lemma or a list-decoding-based amplification, adapted to the time-bounded Kolmogorov complexity setting — for which the author is not aware of an existing construction, or (ii) establishing Theorem Target I directly with a linear gap $\Delta(n) = \Omega(n)$.

Second, option (ii) has a clarifying implication: if $\Delta(n) = \Omega(n)$ is already available from a single instance, no amplification step is needed, since then $N = n$ and $\text{Gap}(x, y) = \Omega(n) = \Omega(N)$ already violates the chain rule outright. Amplification is useful exactly in the regime where the argument above shows it fails, and unnecessary exactly where it would succeed.

It is worth noting that linear gaps of exactly this flavor are not unknown in the literature, which tempers, without resolving, the difficulty just identified. Hu, Manor, and Oliveira prove a failure of *conditional* SoI for rKt with linear error $e(n) = \varepsilon \cdot n$ for an absolute constant $\varepsilon > 0$ [2], in a setting where the conditioning string w is allowed unbounded length, decoupled from $|x| = |y| = n$. This is a different regime from the one needed here: their own ordinary (non-conditional) failure of SoI for rKt comes with only a polylogarithmic error. The existence of a linear-gap result at all, even in a restricted regime, suggests the phenomenon is not structurally forbidden; that it has so far been obtained only under unbounded conditioning, and not in the equal-length regime Revised Target I' needs, is precisely the kind of gap between “known to be possible somewhere nearby” and “established where it is needed” that recurs throughout this program.

Revised Target I'. *Establish Theorem Target I (Section 4), in at least its one-sided form (implication (3)), directly with a linear gap $\Delta(n) = \Omega(n)$; or exhibit a non-trivial amplification scheme — not based on independent direct products — that converts a sublinear per-instance gap into a linear gap at the level of total description length.*

We regard Revised Target I' as strictly harder than the original phrasing of Section 5 suggested, and list it as an explicit open problem in its own right, alongside Target II and the derandomization target of Section 7.

6 Target II: SAT Through Restricted Meta-Complexity

The second obstacle concerns the reduction to conditional K^t .

The Minimum Circuit Size Problem (MCSP) asks whether a given Boolean function has a circuit of size at most s . Whether MCSP is NP-complete remains open. Ilango's 2026 SIAM Journal on Computing paper gives strong evidence for NP-hardness through a random-oracle model and extends the reduction framework to time-bounded Kolmogorov complexity.

Rather than attack full MCSP immediately, introduce a restricted family $\mathcal{F}$ of functions for which the reduction has additional structure.

Theorem Target 2 (Restricted Meta-Complexity Reduction). *There exists an explicit function class $\mathcal{F}$ such that*

$$\text{SAT} \leq_p \text{MCSP}_{\mathcal{F}} \leq_p \text{GapMcK}^t P. \quad (6)$$

The class $\mathcal{F}$ should satisfy three properties:

$\mathcal{F} \subseteq P/\text{poly}$, $\mathcal{F}$ is efficiently samplable, $\mathcal{F}$ admits an efficiently testable structural encoding.

The objective is to retain enough structure for the SAT reduction while avoiding the full derandomization difficulty of unrestricted MCSP.

7 Removing the Random Oracle

Ilango's reduction gives a random-oracle result: with probability one, there is a P/poly reduction from an NP-hard approximation problem to MCSP for circuits with access to a uniformly random oracle, with a corresponding extension to K^t .

The proposed derandomization strategy is not to reproduce the entire random oracle. Instead, construct an explicit family

$$h_n : \{0, 1\}^{a(n)} \rightarrow \{0, 1\}^{b(n)}$$

with the properties required by the reduction.

Derandomization Target. *There exists an efficiently computable pseudorandom family h_n such that replacing the random oracle by h_n preserves the relevant circuit-size gap:*

$$\text{Gap}_{\mathcal{O}}(f) \geq \Delta \implies \text{Gap}_h(f) \geq \Delta - O(\log n). \quad (7)$$

This is weaker than constructing a full deterministic analogue of the random-oracle theorem. It only asks for preservation of the property actually used by the reduction.

8 The Combined Implication — Revisited

Suppose Revised Target I' (Section 5) and Theorem Target II (Section 6) both hold.

From the unconditional failure of symmetry of information for rKt , Revised Target I' yields a family of instances with a deterministic K^t gap that is linear in the total length N :

$$K^t(X, Y) < K^{tc}(X) + K^{tc}(Y \mid X) - \Omega(N). \quad (8)$$

By the exact characterization of Kabanets and Kolokolova (Section 2), (8) refutes the Chain Rule with error $e(N, t) \leq o(N)$ for every admissible function e . Because that Chain Rule is equivalent to the *conjunction* of

- (a) $E \not\subseteq \text{i.o.-SIZE}[2^{o(n)}]$, and
- (b) $\text{GapMcK}^tP \in \text{promise-}P$,

refuting it only yields the disjunction $\neg(a) \vee \neg(b)$ — not $\neg(b)$ on its own. This is precisely where the original version of this section was incomplete: a chain-rule violation is equally consistent with (a) failing instead, in which case E would possess infinitely-often subexponential-size circuits, a separate and strictly weaker statement than anything about NP .

Ruling out this alternative requires a third, previously unstated, ingredient.

Auxiliary Hypothesis (H). $E \not\subseteq \text{i.o.-SIZE}[2^{o(n)}]$.

Hypothesis (H) is a standard assumption in the hardness-versus-randomness literature. It is believed true, is weaker than $P \neq NP$ in the sense that it is not known to follow from it, and its failure would itself be a striking (if not directly contradictory) upper bound on the circuit complexity of E . Whether NP-hardness of GapMcK^tP already forces (H) is, to the author's knowledge, open; if it does, (H) is discharged for free by Target II, and if not, it must be established or assumed separately.

Granting (H), the violation (8) forces $\neg(b)$, i.e., $\text{GapMcK}^tP \notin \text{promise-}P$. Theorem Target II supplies

$$\text{SAT} \leq_p \text{GapMcK}^tP. \quad (9)$$

Therefore, if

$$\text{GapMcK}^tP \in P,$$

then, by (9),

$$\text{SAT} \in P,$$

and hence $P = NP$. Conversely, granting (H), the violation (8), and NP-hardness of $GapMcK^tP$ (needed for (9) to be informative in this direction),

$$\boxed{P \neq NP}.$$

The complete proposed route, with the auxiliary hypothesis made explicit, is therefore

$$\boxed{SAT \longrightarrow MCSP_{\mathcal{F}} \longrightarrow GapMcK^tP \longrightarrow K^t \longleftarrow rKt \quad \text{subject to (H)}}$$

with three bridges requiring proof, not two:

$$\boxed{rKt \longrightarrow K^t \quad (\text{Revised Target I'})}, \quad \boxed{MCSP_{\mathcal{F}} \longrightarrow GapMcK^tP \quad (\text{Target II})}, \quad \boxed{\text{Hypothesis (H)}}.$$

9 Weaker Intermediate Problems

The final theorem need not be attacked first. Several strictly weaker versions are mathematically meaningful, though the weakening in the first item below must now be read in light of Section 5.

First, obtain the local transfer with a gap that is superlogarithmic but sublinear, *together with* a genuine (non-direct-product) amplification scheme — the two must be established jointly, since, as Section 5 shows, the transfer alone no longer suffices on its own once the naive amplification route is closed off.

Second, allow polynomial advice:

$$T \in P/poly.$$

Third, establish the transfer only on average over an efficiently samplable distribution:

$$\Pr_{(x,y) \sim \mathcal{D}_n} [\text{Gap}(x,y) \geq \Delta(n)] \geq 1 - o(1).$$

Fourth, establish the result for restricted circuit models such as bounded-depth circuits or branching programs.

Fifth, determine whether NP-hardness of $GapMcK^tP$ (the target of Section 6) already implies Hypothesis (H) of Section 8, which would remove one of the three bridges entirely.

Any of these results would clarify the precise point at which randomized computational information can or cannot be converted into deterministic computational hardness.

10 Relation to Current Barriers

The proposed program deliberately avoids demanding a direct lower bound against arbitrary polynomial-size circuits. Instead of proving

$$SAT \notin P/poly$$

in one step, it attempts to obtain a hardness witness from computational information and meta-complexity.

This matters because direct circuit lower bounds for unrestricted circuits face longstanding methodological barriers. The meta-complexity formulation can potentially exploit properties of computational descriptions themselves rather than a purely combinatorial property of Boolean functions. It should be said plainly, however, that meta-complexity problems such as MCSP have their own history of resisting hardness proofs, and that Target II inherits this difficulty rather than

avoiding it outright; restricting to a structured family $\mathcal{F}$ narrows the target without guaranteeing tractability.

The 2026 results make this direction attractive nonetheless: the chain rule for time-bounded Kolmogorov complexity now has exact complexity-theoretic characterizations, while randomized symmetry-of-information failure is available unconditionally.

11 Conclusion

We propose a concrete program for proving

$$\boxed{P \neq NP}$$

through computational information. This revision sharpens, rather than resolves, the original proposal. What first appeared to be two independent bridges is, on closer inspection, three:

$$\boxed{rKt \xrightarrow{\text{one-sided, linear gap}} K^t} \quad \boxed{SAT \longrightarrow MCSP_{\mathcal{F}} \longrightarrow GapMcK^tP} \quad \boxed{E \not\subseteq \text{i.o.-SIZE}[2^{o(n)}]}$$

The first exploits the unconditional information-theoretic obstruction already known for rKt , restricted to its lower-bound (one-sided) direction and now understood to require a linear, rather than merely superlogarithmic, gap. The second seeks to replace the random-oracle evidence for meta-complexity hardness by an explicit reduction. The third is a standard, separately-tracked hardness hypothesis on E that the original version of this program did not isolate.

A particularly promising first milestone is no longer, as originally suggested, a transfer gap of merely $\omega(\log n)$ amplified by direct products — Section 5 shows this route to be insufficient. Instead, we recommend the one-sided version of Theorem Target I (implication (3) alone, Section 4) established directly with a linear gap, or a genuinely new combiner-based amplification technique, either of which would constitute independent progress in time-bounded Kolmogorov complexity even without resolving P versus NP .

If all three bridges — the linear-gap transfer, the restricted meta-complexity reduction, and Hypothesis (H) — can ultimately be established, the resulting chain would yield

$$\boxed{SAT \notin P}$$

and therefore

$$\boxed{P \neq NP}.$$

The proposal thus converts the Millennium Problem into three sharply specified mathematical problems, each of whose intermediate weakenings is itself a publishable research target.

12 Outlook: How Hard Is This, Really?

A fair question, raised in independent review of the previous revision, is whether the three bridges identified above are actually within reach, or whether Sections 5 and 8 have merely repackaged the original difficulty of P versus NP under new names. This section attempts an honest calibration, bridge by bridge, rather than treating the corrections above as cosmetic.

On the linear-gap requirement (Revised Target I'). As noted in Section 5, linear-scale failures of symmetry of information are known, but only in a restricted regime (unbounded-length conditioning string) that does not match what the chain-rule violation of Section 8 needs

(comparable-length strings). Closing that distance is a genuine research question, not an engineering step; linear gaps of this kind are typically as hard to obtain as the near-separations they are used to build.

On Hypothesis (H). $E \not\subseteq \text{i.o.-SIZE}[2^{o(n)}]$ belongs to the same family of hypotheses used throughout the hardness-versus-randomness literature to build unconditional pseudorandom generators, and hypotheses of this type have so far resisted unconditional proof for essentially the same reasons that circuit lower bounds in general have resisted proof. It is not a minor bookkeeping addition: showing that Target II’s NP-hardness reduction already forces it, or establishing it on independent grounds, are each open questions of a depth comparable to standard problems in that literature.

On Target II. Restricting the reduction to a structured class $\mathcal{F}$ narrows the object of study, but proving any MCSP-type problem NP-hard directly, rather than through cryptographic assumptions or a random oracle, is a goal that has resisted attack for decades. This is plausibly related to the fact that such problems concern a global property of a function (its circuit size) rather than the local, gadget-based structure that most NP-hardness reductions exploit. Ilango’s random-oracle result is best read as evidence that this obstacle can in principle be crossed, not as a sign that an explicit crossing is imminent.

Taken together. None of the three bridges is close to free; each sits next to a long-standing barrier in complexity theory — pseudodeterministic derandomization for Revised Target I’, hardness-versus-randomness for Hypothesis (H), and the resistance of meta-complexity problems to explicit NP-hardness proofs for Target II. Routes to $P \neq NP$ that avoid a single deep obstacle have a tendency to relocate the difficulty rather than remove it, and this program is unlikely to be an exception. This is not a reason to abandon it: each bridge is a meaningful, well-defined, independently interesting open problem, and the weakened versions of Section 9 remain realistic near-term targets. It is a reason to hold the headline conclusion — a route to $P \neq NP$ — to a low prior, while treating the intermediate mathematics as the program’s genuine, and more likely, payoff.

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